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REVERSAL IN TIME, UNSTABLE STATES, AND PARITY VIOLATION IN ATOMS

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ABSTRACT

The general ideas of symmetry with respect to reversal in time are described with special reference to unstable atomic states. Such symmetry, expected to hold more accurately than parity conservation, excludes by itself some effects which would signal parity violation. The electric dipole moment case is described. Then the Dunford-Lewis-Williams atomic hydrogen set-up is examined from this point of view. A certain signal is excluded in the final strength of the initial state - but not in the final strength of another state, which is what they look at. The way in which the probability balance is maintained, over the various states, is considered explicitly. Even on resonance it is necessary to take account of radiative decay and of non-resonant states.

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1. T, REVERSAL IN TIME¹⁾

Consider, for example, the Schrödinger equation for a spin- $\frac{1}{2}$ particle

$$i\hbar \frac{\partial}{\partial t} \psi(t, \vec{r}) = \left(\frac{\vec{p}^2}{2m} + V + U \vec{\sigma} \cdot \vec{L} \right) \psi(t, \vec{r}) \quad (1)$$

with

$$\vec{p} = (\hbar/i)(\partial/\partial \vec{r}), \quad \vec{L} = \vec{r} \times \vec{p}$$

and where U and V are real functions of $|\vec{r}|$. Complex conjugating and looking at the point $-t$ instead of t

$$i\hbar \frac{\partial}{\partial t} \psi^*(-t, \vec{r}) = \left(\frac{\vec{p}^2}{2m} + V + U \vec{\sigma}^* \cdot \vec{L}^* \right) \psi^*(-t, \vec{r}) \quad (2)$$

If it were not for the spin-orbit complication we would have learned that $\psi^*(-t, \vec{r})$ is a new solution of the old equation. The complication is that with the usual representation

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

we have

$$\sigma_x^* = + \sigma_x \quad \sigma_y^* = - \sigma_y \quad \sigma_z^* = + \sigma_z$$

whereas

$$L_x^* = - L_x \quad L_y^* = - L_y \quad L_z^* = - L_z$$

However, multiply both sides of (2) by $-i\sigma_y$ (the $-i$ is not essential) and note that

$$\sigma_y \sigma_x^* = -\sigma_x^* \sigma_y \quad \sigma_y \sigma_y^* = -\sigma_y^* \sigma_y \quad \sigma_y \sigma_z^* = -\sigma_z^* \sigma_y$$

Then we see that

$$-i\sigma_y \psi^*(-t, \vec{r})$$

is a new solution of the original equation. The substitution

$$\psi(t, \vec{r}) \rightarrow -i\sigma_y \psi^*(-t, \vec{r}) \quad (3)$$

is said to be a symmetry of the equation.

With the new solution the probability $|\psi|^2$ has the same distributions as before but in the reversed time sequence. Expectation values of momentum and spin- and orbital-angular momenta, as well as being reversed in time sequence, are changed in sign. The symmetry is said to be with respect to "reversal in time". Note please that we speak of reversal in time, and not of reversal of time -- whatever that might mean.

Instead of reversing about the time origin, $t = 0$, one can reverse about some other point, say $t = T/2$:

$$\psi(t, \vec{r}) \rightarrow -i\sigma_y \psi^*(T-t, \vec{r}) \quad (4)$$

With time independent U and V in (1), it works for arbitrary T .

With many spin- $\frac{1}{2}$ particles, the generalization is

$$\psi(t, \vec{r}_1, \vec{r}_2, \dots) \rightarrow (-i\sigma_y^1)(-i\sigma_y^2)\dots \psi^*(T-t, \vec{r}_1, \vec{r}_2, \dots) \quad (5)$$

where the spin matrices $-i\sigma_y^1, -i\sigma_y^2, \dots$ act on the spin indices of particles 1, 2, $\dots$. There are corresponding formulae for spins other than $\frac{1}{2}$.

Consider the expansion over a complete set of time-independent basic states

$$\psi(t, \vec{r}) = \sum_n \psi_n(t) \phi_n(\vec{r}) \quad (6)$$

Then the reversed state is

$$\psi'(t, \vec{r}) = \sum_n \psi_n^*(T-t) \phi'_n(\vec{r}) \quad (7)$$

where the ϕ'_n (just $-i\sigma_y \phi_n^*$) are "reversed basic states". It is easily seen in the simple example, and assumed in general, that the reversed basic states are again complete and orthonormal. In Dirac notation, with $|\alpha, t\rangle$ for the original state and $|\alpha', t\rangle$ for the reversed state²⁾,

$$|\alpha, t\rangle = \sum_n \psi_n(t) |n\rangle, \quad |\alpha', t\rangle = \sum_n \psi_n^*(T-t) |n'\rangle \quad (8)$$

Note that for a pair of states

$$\langle \beta', t | \alpha', t \rangle = \langle \alpha, T-t | \beta, T-t \rangle \quad (9)$$

The reversal of order here is characteristic of reversal in time as compared with other symmetries.

The Schrödinger equations in the original and reversed representations are

$$i\hbar d\psi_n(t)/dt = \sum_m \langle n|H|m\rangle \psi_m(t) \quad (10)$$

$$i\hbar d\psi'_n(t)/dt = \sum_m \langle n'|H|m'\rangle \psi'_m(t) \quad (11)$$

For the relation exhibited in (8), i.e.,

$$\psi'_n(t) = \psi_n^*(T-t) \quad (12)$$

we need

$$\langle n'|H|m'\rangle = \langle n|H|m\rangle^* \quad (13)$$

or since H is Hermitian

$$\langle n'|H|m'\rangle = \langle m|H|n\rangle \quad (14)$$

We refer to (13) as the condition for reversibility, and to (14) as the condition for "reciprocity" -- a relation for transition amplitudes to be discussed in a moment.

For an arbitrary (Schrödinger picture) operator A it is useful to define a time-reverse A' by

$$\langle n'|A|m'\rangle = \langle m|A'|n\rangle \quad (15)$$

Then (14) says that the Hamiltonian is even with respect to time reversal:

$$H' = H \quad (16)$$

The position co-ordinate also is even:

$$\vec{r}' = \vec{r} \quad (17)$$

On the other hand, momentum and angular momenta are odd:

$$\vec{p}' = -\vec{p}, \quad \vec{L}' = -\vec{L}, \quad \vec{\sigma}' = -\vec{\sigma} \quad (18)$$

Note that

$$(AB)' = B'A' \quad (19)$$

For,

$$\begin{aligned} \langle n' | AB | m' \rangle &= \sum_{\ell} \langle n' | A | \ell' \rangle \langle \ell' | B | m' \rangle \\ &= \sum_{\ell} \langle m | B' | \ell \rangle \langle \ell | A' | n \rangle = \langle m | B'A' | n \rangle \end{aligned}$$

It follows that

$$(ABC \dots)' = \dots C'B'A' \quad (20)$$

So any power of an even Hamiltonian is even:

$$(H^n)' = (H')^n = H^n \quad (21)$$

It follows that the evolution operator

$$U(T) = \exp(-iHT/\hbar)$$

which transforms the Schrödinger-picture state at zero time to that at time T , is also even:

$$U'(T) = U(T)$$

or

$$\langle n' | U(T) | m' \rangle = \langle m | U(T) | n \rangle \quad (22)$$

-- the fundamental reciprocity relation between transition probability amplitudes.

When Born approximation is good, and the off-diagonal elements of U are just proportional to matrix elements of some interaction Hamiltonian V , (22) just corresponds to the reciprocity of the interaction:

$$\langle n' | V | m' \rangle = \langle m | V | n \rangle \quad (23)$$

When V is Hermitian we have also

$$\langle n' | V | m' \rangle = \langle n | V | m \rangle^* \quad (24)$$

-- giving a relation for transition amplitudes, in Born approximation, not involving exchange of initial and final states.

We will be much concerned with situations involving external electromagnetic fields. For example, in the Hamiltonian operator on the right-hand side of (1) the substitutions

$$\vec{p} \rightarrow \vec{p} - e\vec{A} \quad , \quad V \rightarrow V + e\phi \quad (25)$$

can be made, where $\vec{A}$ and ϕ are external potentials. Then (16) no longer holds. But if in the definition of H' (and other operators) the convention is introduced that external fields as well as operators are "reversed", with

$$\phi'(t, \vec{r}) = \phi(T-t, \vec{r}), \quad \vec{A}'(t, \vec{r}) = -\vec{A}(T-t, \vec{r}) \quad (26)$$

we find

$$H'(t) = H(T-t) \quad (27)$$

This is taken as characteristic in general of reversal in time symmetry with external fields.

With a time dependent H the evolution operator has to be written as a time-ordered product, e.g.,

$$U(T) = \lim_{\delta t \rightarrow 0} T \prod (1 - i\delta t H(t)/\hbar) \quad (28)$$

with a factor for each small interval δt in the complete interval between 0 and T , T signifying that factors for earlier times stand further to the right. Recalling the reversal of order in (20), (28) then implies again (23) -- but with the convention that U' has to be evaluated with reversed fields. Then

$$\langle m' | U(T, \phi', A') | n' \rangle = \langle n | U(T, \phi, A) | m \rangle \quad (29)$$

instead of (22).

2. T_y , REVERSAL IN TIME AND ROTATION THROUGH π

Often a particular direction, which we take as z axis, has a particular role. For example, it can be the direction of a particularly important magnetic field B_z , and/or basic states may be chosen with definite angular momentum projections J_z . Then it is convenient to define an operation related to time reversal but leaving B_z and J_z unchanged. It is obtained by following T by a rotation through an angle $-\pi$ about (say) the y axis, and we label it T_y .

The rotation of a spinor through π about the y axis requires multiplication by $+i\sigma_y$, so instead of (14)

$$\psi(t, x, y, z) \rightarrow \psi^*(t, -x, y, -z) \quad (30)$$

All the developments of the last section go through with T_y instead of T . In particular (29) again holds, with, instead of (26)

$$\begin{aligned} \phi'(t, x, y, z) &= + \phi(T-t, -x, y, -z) \\ A'_x(t, x, y, z) &= + A_x(T-t, -x, y, -z) \\ A'_y(t, x, y, z) &= - A_y(T-t, -x, y, -z) \\ A'_z(t, x, y, z) &= + A_z(T-t, -x, y, -z) \end{aligned} \quad (31)$$

The components of $\vec{B}$ go in the same way as those of $\vec{A}$, and those of $\vec{E}$ with opposite signs.

The operation T_y , like T , changes a state with exponential time dependence into another

$$\exp(-iEt) \rightarrow \exp(-iE^*t) \quad (32)$$

In the case of a stable state (with $E = E^*$) which is not degenerate (except possibly with one of different J_z) the T_y reversed state must be the same as the original -- except possibly for a phase factor. So, by incorporating a suitable phase factor (which is complex-conjugated by T_y) in the definition of $|n\rangle$,

$$|n'\rangle = |n\rangle \quad (33)$$

in the case of a non-degenerate stable state. An operator A which is even or odd under T_y has matrix elements between two such states with

$$\langle m | A | n \rangle = \pm \langle n | A | m \rangle \quad (34)$$

If A is also Hermitian

$$\langle m | A | n \rangle = \pm \langle m | A | n \rangle^* \quad (35)$$

i.e., the matrix elements are real and imaginary for even and odd operators, respectively.

It should be noted that the phase convention of (33) is not quite that established by Condon and Shortly. It is rather that used (for example) by Landau and Lifshitz. In the single particle spherical wave functions, for example,

$$Y_{\ell m}(\theta, \phi) R_{\ell}(r)$$

(with real radial functions R) an extra factor i^{ℓ} (as compared with Condon and Shortly) is included in the spherical harmonics)

$$Y_{\ell m} = (-1)^{(m+|m|)/2} i^{\ell} \left[\frac{(\ell-|m|)!}{(\ell+|m|)!} \frac{2\ell+1}{4\pi} \right]^{1/2} P_{\ell}^{|m|}(\cos \theta) e^{im\phi} \quad (36)$$

3. UNSTABLE STATES

We are concerned with atomic states that are unstable against photon emission. But we are mainly interested here in processes in which photon emission happens not to occur. We restrict attention then to the zero-photon projection of the total state. To avoid explicit reference to the photon sector an "effective" Hamiltonian can be introduced which allows for radiation reaction³⁾. The effective Hamiltonian is not Hermitian in general, for the norm of the no-photon sector is not separately conserved -- probability escapes into the radiative sector. However, when T or T_y is a symmetry the effective Hamiltonian will respect the appropriate reciprocity relation (27), for this is implied by the reciprocity (29) of transition amplitudes -- which holds in particular for those in the zero-photon sector. The non-Hermitian effective Hamiltonian need not, however, respect a reversibility condition like (13). Indeed the reversal operation (3) converts a decaying state into a growing state, respecting advanced rather than retarded boundary conditions and outside the competence of the effective Hamiltonian -- and not of direct physical interest.

In what follows the Hamiltonian will always be the effective Hamiltonian, and states will belong to the non-photon sector -- unless the contrary is stated.

The eigenstates $|\alpha\rangle$ of the effective Hamiltonian, defined by

$$(E_{\alpha} - H)|\alpha\rangle = 0 \quad (37)$$

have simple time dependence:

$$|\alpha\rangle \rightarrow e^{-iE_{\alpha}t} |\alpha\rangle$$

The E_{α} are in general complex, with

$$- \operatorname{Im} E_{\alpha} = \Gamma_{\alpha}/2 \gg 0 \quad (38)$$

so that (38) corresponds to decreasing norm.

With H non-Hermitian the different eigenstates are in general not orthogonal. In a superposition

$$|f\rangle = \sum f_{\alpha} |\alpha\rangle$$

then

$$f_{\alpha} \neq \langle \alpha | f \rangle$$

It is useful to introduce a new kind of conjugate, $\langle \bar{\alpha} |$, such that

$$f_{\alpha} = \langle \bar{\alpha} | f \rangle \quad (39)$$

This requires

$$\langle \bar{\alpha} | \beta \rangle = \delta_{\alpha\beta}$$

These new conjugates can be identified with the "left eigenstates" defined by

$$\langle \bar{\beta} | (\bar{E}_{\beta} - H) = 0 \quad (40)$$

They are not just the ordinary conjugates $\langle \beta |$ of the "right eigenvectors" defined by (37), for the latter would satisfy rather

$$\langle \beta | (E_{\beta}^* - H^+) = 0 \quad (41)$$

Multiplying (37) on the left by $\langle \bar{\beta} |$ or (40) on the right by $|\alpha\rangle$,

$$\langle \bar{\beta} | \alpha \rangle (\bar{E}_{\beta} - E_{\alpha}) = 0 \quad (42)$$

so either

$$\langle \bar{\beta} | \alpha \rangle = 0 \quad \text{or} \quad \bar{E}_{\beta} = E_{\alpha} \quad (43)$$

Assume now that there is no degeneracy, or at most that between different members of a rotational multiplet with different J_z quantum numbers. Then for each $|\alpha\rangle$ there is at most one $\langle \bar{\beta} |$, and for each $\langle \bar{\beta} |$ at most one $|\alpha\rangle$, for which $\langle \bar{\beta} |^{\alpha} \neq 0$. Assume that there is one such, in each case, and denote corresponding states by $(|\alpha\rangle, \langle \bar{\alpha}|)$, etc. Then normalizing $\langle \bar{\alpha} |$ so that

$$\langle \bar{\alpha} | \alpha \rangle = 1 \quad (44)$$

we have

$$\langle \bar{\beta} | \alpha \rangle = \delta_{\beta\alpha} \quad (45)$$

It is useful to note the form taken here by first order perturbation theory.
With

$$H = H_0 + X \quad (46)$$

and using round brackets to denote unperturbed states, multiplication of (40) on the right by $|\alpha\rangle$ and (37) on the left by $\langle \bar{\beta}|$ gives to first order in X :
with $\alpha = \beta$

$$E_{\alpha} - E_{\alpha}^0 = (\bar{\alpha} | X | \alpha) \quad (47)$$

and with $\alpha \neq \beta$

$$\langle \bar{\beta} | \alpha \rangle = - \langle \bar{\beta} | \alpha \rangle = (\bar{\beta} | X | \alpha) / (E_{\alpha} - E_{\beta}) \quad (48)$$

When T_y is a symmetry $\langle \bar{\alpha}|$ and $\langle \alpha'|$ can be identified up to a factor:

$$\langle \bar{\alpha} | = \langle \alpha' | \langle \alpha' | \alpha \rangle^{-1} \quad (49)$$

To see this multiply (37) on the left by a complete set of states $\langle m|$ and use (14) and that the set $|m'\rangle$ is also complete, obtaining

$$\langle \alpha' | (E_{\alpha} - H) = 0 \quad (50)$$

in agreement with (40).

When the system is disturbed⁴⁾ by external fields transitions will occur between the states $|\alpha\rangle$:

$$|\alpha\rangle \rightarrow U|\alpha\rangle = \sum_{\beta} |\beta\rangle U_{\beta\alpha} + \dots \quad (51)$$

where ... indicates radiative states. From (39) the transition amplitude is

$$U_{\beta\alpha} = \langle \bar{\beta} | U | \alpha \rangle \quad (52)$$

When T_y is a symmetry (49) and (29) give

$$\begin{aligned} \langle \beta' | \beta \rangle \langle \bar{\beta} | U(\phi', A') | \alpha \rangle &= \langle \beta' | U(\phi', A') | \alpha \rangle \\ &= \langle \alpha' | U(\phi, A) | \beta \rangle = \langle \alpha' | \alpha \rangle \langle \bar{\alpha} | U(\phi, A) | \beta \rangle \end{aligned}$$

or

$$\langle \beta' | \beta \rangle U_{\beta\alpha}(\phi', A') = \langle \alpha' | \alpha \rangle U_{\alpha\beta}(\phi, A) \quad (53)$$

In particular

$$U_{\alpha\alpha}(\phi', A') = U_{\alpha\alpha}(\phi, A) \quad (54)$$

i.e., the diagonal amplitude is even with respect to reversal of the fields.

4. ELECTRIC DIPOLE MOMENT

The question of electric dipole moment⁵⁾⁻⁹⁾ can be interpreted in various ways. One can ask whether there is a linear Stark effect (in analogy with magnetic Zeeman effect), or one can ask whether polarization processes in an electric field, or one can ask whether an electric field exerts a torque on the system.

The first two questions are equivalent, for polarization precession is due to systematic phase drift between states of different J_z in a coherent superposition, arising from field induced energy differences. The first order shift in energy E_α of state $|\alpha\rangle$ in a uniform electric field E_z is from (47)

$$\delta E_\alpha = E_z \langle \bar{\alpha} | D_z | \alpha \rangle \quad (55)$$

where D_z is the electric dipole moment operator. From (49)

$$\langle \bar{\alpha} | D_z | \alpha \rangle = \langle \alpha' | D_z | \alpha \rangle / \langle \alpha' | \alpha \rangle \quad (56)$$

From (15)

$$\langle \alpha' | D_z | \alpha \rangle = \langle \alpha' | D'_z | \alpha' \rangle \quad (57)$$

$$= \langle \alpha' | D_z | \alpha \rangle \quad (58)$$

But D_z is odd with respect to T_y [as follows in general from (27) and the oddness of E_z in (31)]:

$$D'_z = -D_z \quad (59)$$

So

$$\langle \bar{\alpha} | D_z | \alpha \rangle = 0 \quad (60)$$

Thus as regards energy shifts and polarization precession, the electric dipole moment is forbidden by T_y symmetry -- and therefore by reversal in time and rotation symmetries combined^{6),7)}.

The above argument from T_y works even in the presence of a magnetic field B_z , which does not spoil T_y although it spoils T and rotation symmetry separately. In the absence of such a field, and with full rotation symmetry (neglecting the weak E_z) a stronger conclusion holds. Not only do the diagonal matrix elements of D_z vanish, but so do all matrix elements of all components of D within a given rotational multiplet, for they differ only by Clebsch-Gordan coefficients.

The torque exerted by the field $\vec{E}$ on the system is

$$\vec{E} \times \langle \alpha | \vec{D} | \alpha \rangle \quad (61)$$

It involves the ordinary expectation value

$$\langle \alpha | \vec{D} | \alpha \rangle \quad (62)$$

rather than

$$\langle \bar{\alpha} | \vec{D} | \alpha \rangle \quad (63)$$

In the unstable case, with $\langle \bar{\alpha} | \neq \langle \alpha |$, T_y does not require (62) to vanish, and in general (with parity non-conservation) it does not vanish, as observed by Zeldovich⁵⁾. Then angular momentum is delivered to the system at a non-zero rate (61). Where does it go? The angular momentum of the undecayed state does not precess. However, the norm in this sector diminishes and the norm in the decay sector increases. If there is a difference of angular momentum per unit norm in the two sectors, the angular momentum of the system as a whole changes steadily. It has been confirmed explicitly in a simple case that this is what happens^{8),9)}. The angular momentum per unit norm in the decay sector is tilted a little with respect to that in the parent sector, and the transfer of norm between the two sectors accounts for the rate (61) of delivery of angular momentum.

Both (62) and (63) must vanish when parity is conserved, for $\vec{D}$ is odd with respect to space inversion. Consider a system in which both T_y and P are respected, and add a weak interaction X which violates P but not T_y . It is instructive to see in explicit perturbation theory how (63) continues to vanish, but not (62). From (48), to first order in X

$$\langle \bar{\alpha} | D_z | \alpha \rangle = \sum_{\beta} \left\{ \frac{(\bar{\alpha} | D_z | \beta)(\bar{\beta} | X | \alpha)}{E_{\alpha} - E_{\beta}} + \frac{(\bar{\alpha} | X | \beta)(\bar{\beta} | D_z | \alpha)}{E_{\alpha} - E_{\beta}} \right\} \quad (64)$$

whereas

$$\langle \alpha | D_z | \alpha \rangle = \sum_{\beta} \left\{ \frac{(\alpha | D_z | \beta)(\bar{\beta} | X | \alpha)}{E_{\alpha} - E_{\beta}} + \frac{(\bar{\beta} | X | \alpha)^* (\beta | D_z | \alpha)}{E_{\alpha}^* - E_{\beta}^*} \right\} \quad (65)$$

The cancellation of the two terms in (64) follows readily from the oddness and evenness of D_z and X , respectively with respect to T_y :

$$\begin{aligned} (\bar{\alpha} | D_z | \beta) &= (\alpha' | D_z | \beta) / (\alpha' | \alpha) \\ &= -(\beta' | D_z | \alpha) / (\alpha' | \alpha) = -(\bar{\beta} | D_z | \alpha)(\beta' | \beta) / (\alpha' | \alpha) \end{aligned} \quad (66)$$

$$\begin{aligned} (\bar{\beta} | X | \alpha) &= (\beta' | X | \alpha) / (\beta' | \beta) \\ &= +(\alpha' | X | \beta) / (\beta' | \beta) = +(\bar{\alpha} | X | \beta)(\alpha' | \alpha) / (\beta' | \beta) \end{aligned} \quad (67)$$

This does not work at all for (65).

5. DUNFORD, LEWIS, AND WILLIAMS¹⁰⁾

They consider a parity forbidden electric dipole transition between two S states of atomic hydrogen. The two states are of different energy, so the discussion of the preceding section does not apply.

To enhance parity admixture, by bringing close together the energies of the final S state and of certain P states, a strong magnetic field $\vec{B}$ is applied. A weak static electric field $\vec{E}_s$, perpendicular to $\vec{B}$, provides a deliberate and variable parity admixture. An alternating electric field induces the transition:

$$\vec{E}(t) = \vec{\epsilon} e^{-i\omega t} + \vec{\epsilon}^* e^{i\omega t} \quad (68)$$

Assuming that only the first term in (68) is nearly resonant, there will be terms (among others) proportional to

$$\vec{\epsilon} \cdot \vec{E}_s \quad \text{and} \quad \vec{\epsilon} \cdot \hat{B} \quad (69)$$

in the transition amplitude, and a corresponding interference term in the transition probability

$$\text{Re } C \vec{\epsilon}^* \cdot \vec{E}_s \vec{\epsilon} \cdot \vec{B} \quad (70)$$

where C is some coefficient dependent on ω and $|\vec{B}|$ but not $\vec{\epsilon}$ or $\vec{E}_s$. This is the hoped for parity violation signal. Does symmetry with respect to reversal in time have anything to say about it?

Take $\vec{B}$ and $\vec{E}_s$ in the z and y directions, respectively; then both are invariant under the T_y transformation (31). For the alternating field

$$\begin{aligned} E'_y(t) &= \epsilon_y e^{-i\omega(\tau-t)} + \epsilon_y^* e^{i\omega(\tau-t)} \\ &= \epsilon'_y e^{-i\omega t} + \epsilon'^*_y e^{i\omega t} \end{aligned} \quad (71)$$

with

$$\epsilon'_y = \epsilon_y^* e^{i\omega\tau}$$

In the same way

$$\epsilon'_z = -\epsilon_z^* e^{i\omega\tau} \quad (72)$$

For DLR, ϵ_y and ϵ_z are in phase, so

$$\epsilon_y \epsilon_z^* = \epsilon_y^* \epsilon_z \quad (73)$$

Then (70), because of the minus in (72), is seen to be odd with respect to T_y .

So, from (54), T_y forbids the term (70) in the squared amplitude

$$|\langle \bar{\alpha} | U | \alpha \rangle|^2 \quad (74)$$

for the system to remain in initial state $|\alpha\rangle$. This observation was made to me by J. Hufnagle, R.T. Robiscoe, and V.L. Telegdi. There is no similar conclusion for the transition probability to another state $|\beta\rangle$,

$$|\langle\bar{\beta}|V|\alpha\rangle|^2 \quad (75)$$

-- for then T_y simply relates different transitions, $\alpha \rightarrow \beta$ and $\beta \rightarrow \alpha$. This is so even in Born approximation [see (24)] when $\langle\bar{\beta}| \neq \langle\beta|$. But then the following puzzle arises. Can one not, by well satisfying the resonance condition, ensure that $|\beta\rangle$ is the only significant final state? Should not (74) and (75) then add to unity? And how could that be consistent with the presence of a term (70) in (75) but not (74)?

To dispose of this puzzle we have to go in more detail into the theory of induced transitions involving unstable states.

Denote by V the time-dependent perturbation

$$V(t) = \vec{\epsilon} \cdot \vec{D} e^{-i\omega t} + \text{c.c.} \quad (76)$$

$$= V_{\omega} e^{-i\omega t} + V_{-\omega} e^{i\omega t} \quad (77)$$

Let the unperturbed Hamiltonian (containing $\vec{E}_s$ and $\vec{B}$) give rise to quasi-steady states $|n\rangle$. The coefficients f in the superposition

$$\sum_n f_n |n\rangle \quad (78)$$

satisfy

$$(i\hbar d/dt - E_n) f_n = 0 \quad (79)$$

in the unperturbed case. The perturbed equations, with $\hbar = 1$, are

$$(id/dt - E_n) f_n = \sum_m \langle n|V|m\rangle f_m \quad (80)$$

These we solve by successive approximation. Writing

$$V = \sum_{\omega} V_{\omega} e^{-i\omega t} \quad (81)$$

and substituting on the right in (80) the zero order

$$f_m(t) = f_m(0) e^{-i E_m t} \quad (82)$$

we find in first order

$$f_n(t) = e^{-i E_n t} f_n(0) + \sum_{\omega} \sum_m \left(e^{-i(\omega + E_m)t} - e^{-i E_n t} \right) \frac{\langle \bar{n} | V_{\omega} | m \rangle}{\omega + E_m - E_n} f_m(0) \quad (83)$$

To simplify things in a relevant way assume:

$$f_{\alpha}(0) = 1, \quad f_n(0) = 0 \text{ for } n \neq \alpha \quad \left. \vphantom{\sum_{\omega} \sum_m} \right\} \quad (84)$$

$$\text{Just one frequency, } \omega, \text{ gives rise to important terms, the others giving too-large energy denominators} \quad \left. \vphantom{\sum_{\omega} \sum_m} \right\} \quad (85)$$

$$\text{For just one state, } |\beta\rangle, \text{ is the resonance condition well met} \quad \left. \vphantom{\sum_{\omega} \sum_m} \right\} \quad (86)$$

$$|\omega + E_{\alpha} - E_{\beta}| t \ll 1$$

for t of interest.

$$\text{For states } |n\rangle \text{ other than } |\alpha\rangle \text{ and } |\beta\rangle, \quad \left. \vphantom{\sum_{\omega} \sum_m} \right\} \quad (87)$$

$$(-\text{Im } E_n) t = \Gamma_n t / 2 \gg 1$$

for typical t of interest.

$$\text{The initial state } |\alpha\rangle \text{ is stable:} \quad \left. \vphantom{\sum_{\omega} \sum_m} \right\} \quad (88)$$

$$\Gamma_{\alpha} = 0, \quad \langle \bar{\alpha} | = \langle \alpha |, \quad \langle \alpha | n \rangle = 0 \text{ for } n \neq \alpha$$

$$\text{The perturbation does not connect } |\alpha\rangle \text{ to itself:} \quad \left. \vphantom{\sum_{\omega} \sum_m} \right\} \quad (89)$$

$$\langle \bar{\alpha} | V_{\omega} | \alpha \rangle = 0$$

Then the important terms in (83) are

$$\left. \begin{aligned} f_{\alpha}(t) &= e^{-i E_{\alpha} t} \\ f_{\beta}(t) &= e^{-i(\omega + E_{\alpha})t} \langle \bar{\beta} | V_{\omega} | \alpha \rangle (-it) \\ f_n(t) &= e^{-i(\omega + E_{\alpha})t} \langle \bar{n} | V_{\omega} | \alpha \rangle / (\omega + E_{\alpha} - E_n) \end{aligned} \right\} \quad (90)$$

-- the latter for n other than α or β . Substituting back into (80) gives a second approximation, from which we note only the correction to f_α :

$$f_\alpha(t) = e^{-iE_\alpha t} \left(1 - \frac{t^2}{2} \langle \bar{\alpha} | V_{-\omega} | \beta \rangle \langle \bar{\beta} | V_\omega | \alpha \rangle - it \sum_{n \neq \beta} \frac{\langle \bar{\alpha} | V_{-\omega} | n \rangle \langle \bar{n} | V_\omega | \beta \rangle}{\omega + E_\alpha - E_n} \right) \quad (91)$$

The norm of the final state, in so far as it is accounted for by the f 's, is not unity. In particular, the only fully resonant terms, proportional to t^2 , are from $|f_\alpha|^2$

$$- t^2 \operatorname{Re} \langle \bar{\alpha} | V_{-\omega} | \beta \rangle \langle \bar{\beta} | V_\omega | \alpha \rangle \quad (92)$$

and from $|f_\beta|^2$

$$+ t^2 \langle \bar{\beta} | V_\omega | \alpha \rangle^* \langle \bar{\beta} | V_\omega | \alpha \rangle \quad (93)$$

These do not cancel. Although $V_\omega = (V_{-\omega})^\dagger$, and $\langle \bar{\alpha} | = \langle \alpha |$,

$$\langle \bar{\beta} | \neq \langle \beta | \quad (94)$$

How then is probability conserved? We have to take account of the radiative sector, whose existence is responsible for the instability of the states, and for (94).

The rate of radiative decay is quickly obtained by computing from (79) the rate at which the norm of (78) decreases (for the Hermitian V does not change this rate):

$$\sum_{n,m} i(E_n - E_m^*) f_m^* f_n \langle m | n \rangle \quad (95)$$

Standard unitarity amplitudes¹¹⁾ relate this directly to decay amplitudes, in a way exhibiting positive definiteness. For the present purpose it is conveniently rewritten

$$2 \operatorname{Re} \sum_{n,m} i(E_\beta - E_m^*) f_m^* f_n \langle m | n \rangle \quad (96)$$

-- where we have neglected Γ_β in comparison with other Γ 's, as already in assumptions (3) and (4) above. Substituting f 's from (90), picking out the term linear in t (an interference term between states $|\beta\rangle$ and $|n\rangle$), and integrating over time, the t^2 term in the radiative transition probability is

$$E^2 \operatorname{Re} \sum_{m \neq \beta} \langle m | \beta \rangle \langle \bar{m} | V_\omega | \alpha \rangle^* \langle \bar{\beta} | V_\omega | \alpha \rangle \quad (97)$$

Adding this to (93), and noting the completeness relation

$$\left(\langle \beta | \beta \rangle \langle \bar{\beta} | V_\omega | \alpha \rangle + \sum_{m \neq \beta} \langle \beta | m \rangle \langle \bar{m} | V_\omega | \alpha \rangle \right)^* \left. \vphantom{\sum_{m \neq \beta}} \right\} \quad (98)$$

$$= \left(\langle \beta | V_\omega | \alpha \rangle \right)^* = \langle \alpha | V_{-\omega} | \beta \rangle$$

we obtain finally the negative of (92). Even on resonance, then, radiative final states are important in the probability balance. A T_y -odd term in (97) can cancel with one in (93), leaving none in (92). Non-radiative intermediate states other than β are also important, because of the interference origin of (97).

Finally, to arrive at more practical formulae, let us treat by the first order perturbation theory of (48) the parity violating interaction

$$X + E_s D_y \quad (99)$$

Then

$$\langle \bar{\beta} | V_\omega | \alpha \rangle = \sum_n (\bar{\beta} | X + E_s D_y | n) (\bar{n} | \vec{\epsilon} \cdot \vec{D} | \alpha) / (E_\beta - E_n) \quad (100)$$

Now introduce for the specific states of Dunford, Lewis, and Williams the labels of Lamb: α_0 and β_0 for the initial and final S states, and e_0 and e_1 for the important nearby P states, with the numerical suffix indicating the z component of angular momentum. Allowing for rotational selection rules

$$\langle \bar{\beta}_0 | V_\omega | \alpha_0 \rangle = \frac{(\bar{\beta}_0 | X | e_0) (e_0 | \epsilon_z D_z | \alpha_0)}{E(\beta_0) - E(e_0)} + \frac{(\bar{\beta}_0 | E_s D_y | e_1) (e_1 | \epsilon_y D_y | \alpha_0)}{E(\beta_0) - E(e_1)} \quad (101)$$

$$= i F \epsilon_z / \Delta_0 + G E_s \epsilon_y / \Delta_1 \quad (102)$$

-- where the Δ 's are the energy denominators and this defines F and G. The radiative widths of the e levels are very important in the denominators, being comparable with the real parts of the splittings. But radiative corrections give only small effects in the numerators; we ignore them there, taking, in particular,

$$\langle \bar{\beta}_0 | = \langle \beta_0 |$$

Then from (35) the matrix elements of X and D_y (even under T_y) are real, while that of D_z (odd) is imaginary. So F and G in (102) are real.

In the same way

$$\langle \alpha_0 | V_{-\omega} | \beta_0 \rangle = -iF \epsilon_z^* / \Delta_0 + G E_s \epsilon_y^* / \Delta_1 \quad (103)$$

Note that between (103) and (102) there is no complex conjugation of energy denominators.

Multiplying (102) and (103) we find for the term of type (70) in (92)

$$-t^2 \operatorname{Re} (\Delta_0 \Delta_1)^{-1} (\epsilon_z \epsilon_y^* - \epsilon_z^* \epsilon_y) iFG E_s$$

Because of (73) this vanishes, in accord with the general reasoning. But, squaring the modules of (102) we find in the transition probability (93), the term

$$t^2 \operatorname{Re} (\Delta_0 \Delta_1^*)^{-1} \epsilon_y^* \epsilon_z iFG E_s$$

-- which does not vanish in general, even with $\epsilon_y^* \epsilon_z$ real, when the Δ 's are complex.

Note that in this account we included $E_s D_y$ in the time-independent Hamiltonian, in effect assuming $\vec{E}_s$ present during preparation of initial state $|\alpha\rangle$. If E_s is actually switched on and off with the alternating field this can be taken account of by transferring $E_s D_y$ to the perturbation V above. Alternatively, one can add to initial and final states small terms correcting our $|\alpha\rangle$ and $|\beta\rangle$. But the extra states would not satisfy the resonance condition (86), and not change the resonant term (93), while the renormalization of the original states would be an effect of higher order in E_s . So we do not obtain a different result for the resonant term when $E_s D_y$, and even X , is regarded as switched on and off with the alternating field.

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- 2) It might be better to put the prime outside the bracket, i.e., $|\alpha, t\rangle'$ and $|n\rangle'$, for we are changing the state associated with fixed labels, as is clear in (7) with ϕ_n' and not ϕ_n . A similar remark applies in connection with (39) where it might be better to write $\langle\beta|$ rather than $\langle\bar{\beta}|$.
- 3) See, for example, N. Byers, S. MacDowell and C.N. Yang, in Seminar on High-Energy Physics and Elementary Particles, Trieste (1965) 953.
- 4) It is tacitly assumed here that a limited number of unstable states, not too different in energy²⁾, form a sufficiently complete set for the purpose at hand.
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- 12) When such a superposition is normalized it is tempting to refer to f_α as the "probability amplitude for state $|\alpha\rangle$ ", and such terminology is not entirely avoided in this paper. It has be used with some caution. With the $|\alpha\rangle$'s not orthogonal the norm involves interference terms, and $\sum |f_\alpha|^2 \neq 1$. The various states $|\alpha\rangle$ are not to be thought of as mutually exclusive possibilities. However, if one has some kind of filter set-up eliminating states other than $|\alpha\rangle$, then $|f_\alpha|^2$ is indeed the transmission probability.